Human-Centric Multi-Agent Control (HC-MARL)

Corrected and Complete Mathematical Framework

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1 Executive Summary

This document presents the complete, line-by-line verified mathematical architecture for the Human-Centric Multi-Agent Reinforcement Learning (HC-MARL) framework. The following corrections have been implemented relative to previous iterations:

C1. MMICRL objective: The weighted entropy objective $\lambda_1 H[\pi(\tau)] - \lambda_2 H[\pi(\tau \mid z)]$ is not equivalent to mutual information $I(\tau; z)$ unless $\lambda_1 = \lambda_2$. The objective is now correctly characterised as a weighted information-theoretic trade-off with an explicit residual term.

C2. ECBF switched-mode proof: The flawed first-derivative argument is replaced by a direct Nagumo invariance proof on $h(x) = \Theta_{\max} - M_F \geq 0$, yielding an explicit and verifiable design requirement $\Theta_{\max} \geq F/(F + Rr)$ for each muscle group.

C3. Dual safety barrier: A second barrier function $h_2(x) = M_R \geq 0$ is introduced to enforce physiological validity of the resting compartment, preventing transient recruitment from an empty pool.

C4. Nash Social Welfare: The centralised allocation objective $\max_{\{a_{ij}\}} \sum_i \ln(U(i, j^*(i)) - D_i)$ is correctly identified as maximising the Nash Social Welfare Function (NSWF), not "Nash Bargaining" in the game-theoretic sense. The axiomatic connection to Nash's framework is preserved through the NSWF's equivalence to the Nash Bargaining Solution under specific conditions.

C5. Action-to-neural-drive interface: A new section specifies the mapping from discrete RL task assignments to continuous neural drive trajectories $C(t)$, closing the gap between the allocation layer and the physiological model.

Preserved results. The 3CC-r ODE system, mass conservation, sustainability bound $\delta_{\max} = R/(F + R)$, parameter calibration, ECBF relative-degree-2 identification, and divergent disagreement utility $D_i(M_F) = \kappa \cdot (M_F)^2/(1 - M_F)$ were verified correct and are retained without change.

2 Notation and Conventions

Symbol	Definition
$M_R(t)$, $M_A(t)$, $M_F(t)$	Fraction of motor units in Resting, Active, and Fatigued compartments at time t. Each lies in $[0, 1]$.
$x(t) = [M_R, M_A, M_F]^\top$	Physiological state vector in $\mathbb{R}^3$.
$C(t)$	Neural drive (recruitment rate), $C(t) \in [0, C_{\text{cap}}]$, units min^{-1}.
F	Fatigue rate constant, units min^{-1}. Muscle-specific.
R	Base recovery rate constant, units min^{-1}. Muscle-specific.
r	Dimensionless reperfusion multiplier ($r > 1$).
$R_{\text{eff}}(t)$	Effective recovery rate: R during work, $R \cdot r$ during rest.
$T_L(t)$	Target load (fraction of MVC demanded by current task).
δ	Duty cycle $= M_A$ at steady state (dimensionless).
$\Theta_{\max}$	Maximum allowable fatigue fraction (safety threshold).
$\Theta_{\max}^{\min}$	Rest-phase safety threshold $= F/(F + Rr)$ (Def. 5.4).
$h(x)$, $h_2(x)$	Barrier functions for fatigue ceiling and resting floor.
ψ_0, ψ_1	ECBF composite barriers (degree 0 and 1).
α_1, α_2	Positive ECBF gain parameters (class-$\mathcal{K}$ [25]). Used only in Sections 5–5.4.
α_3	CBF gain for the resting-pool barrier h_2.
λ_1, λ_2	MMICRL weighting coefficients (Section 4).
z	Latent variable denoting agent type in MMICRL.
τ	Trajectory $\tau = (s_0, a_0, s_1, a_1, \ldots)$. Used only in Section 4.
a_{ij}	Binary assignment variable: $a_{ij} = 1$ iff task $j \in \mathcal{J}$ assigned to worker $i \in \mathcal{I}$. Used only in Section 6.
$U(i, j)$	Productivity utility of worker i performing task j.
$D_i(M_F)$	Disagreement utility (value of the outside option: resting).
κ	Positive scaling constant for disagreement utility.
$\mathcal{I} = \{1, \ldots, N\}$	Set of N human workers.
$\mathcal{J} = \{1, \ldots, M\}$	Set of M productive tasks in the current allocation round.
$\mathcal{J}_0 = \mathcal{J} \cup \{0\}$	Augmented task set including the rest option (task 0).
ϵ	Small positive constant for rest-task surplus (Eq. 31).

3 Physiological Control: The 3CC-r Model

We model each human worker as a dynamic biological plant governed by mass-action kinetics. The compartmental structure was introduced by Liu, Brown & Yue [1], extended to submaximal tasks by Xia & Frey-Law [2], and augmented with a reperfusion multiplier by Looft, Herkert & Frey-Law [4]. The empirical foundation for muscle endurance modelling traces to the pioneering endurance-time measurements of Rohmert [26], whose data underpin all subsequent calibration efforts [6, 3].

3.1 State Space and Conservation Law

Definition 3.1 (State Space)**.** *The state vector $x(t) \in \mathbb{R}^3$ represents the fraction of motor units in three compartments [1]:*

- $M_R(t)$*: Resting / Recovered.*
- $M_A(t)$*: Active / Generating force.*
- $M_F(t)$*: Fatigued / Metabolically refractory.*

Theorem 3.2 (Conservation of Mass)**.** *The total motor unit fraction is conserved [1]:*

$$M_R(t) + M_A(t) + M_F(t) = 1, \quad \forall\, t \geq 0. \tag{1}$$

Equivalently, $\frac{dM_R}{dt} + \frac{dM_A}{dt} + \frac{dM_F}{dt} = 0$.

Proof. Summing the three ODEs in (2)–(4):

$$\big(C - FM_A\big) + \big(FM_A - R_{\text{eff}} M_F\big) + \big(R_{\text{eff}} M_F - C\big) = 0. \qquad \square$$

3.2 System Dynamics (ODEs)

The flow between compartments is governed by the neural drive $C(t)$, the fatigue rate F, and the effective recovery rate $R_{\text{eff}}(t)$. Recovery routes from $M_F \to M_R$ (not directly to M_A), following the correction of Xia & Frey-Law [2] over Liu et al. [1].

$$\frac{dM_A}{dt} = C(t) - F \cdot M_A(t), \tag{2}$$

$$\frac{dM_F}{dt} = F \cdot M_A(t) - R_{\text{eff}}(t) \cdot M_F(t), \tag{3}$$

$$\frac{dM_R}{dt} = R_{\text{eff}}(t) \cdot M_F(t) - C(t). \tag{4}$$

Remark 3.3 (Physical Validity Constraints)**.** *Beyond conservation* (1)*, the state must satisfy $M_R, M_A, M_F \in [0, 1]$ for physiological validity [1, 2]. The constraint $M_R \geq 0$ (equivalently $M_A + M_F \leq 1$) is not automatically enforced by the ODEs under arbitrary control inputs. This motivates the dual barrier system in Section 5.*

3.3 The Reperfusion Switch

To capture the rapid vasodilation upon cessation of effort [4], we define $R_{\text{eff}}(t)$ as a switched parameter:

$$R_{\text{eff}}(t) = \begin{cases} R & \text{if } T_L(t) > 0 \quad \text{(Work phase: standard recovery)}, \\ R \cdot r & \text{if } T_L(t) = 0 \quad \text{(Rest phase: reperfusion)}. \end{cases} \tag{5}$$

Here R is the base recovery rate (min^{-1}) and $r > 1$ is the dimensionless reperfusion multiplier. Optimal values were determined by Looft et al. [4] via meta-analysis of 63 publications: $r = 15$ for shoulder/back/ankle/knee/elbow, $r = 30$ for grip. Shoulder $r = 15$ was independently validated by Looft & Frey-Law [5].

3.4 Maximum Sustainable Duty Cycle

Theorem 3.4 (Sustainability Bound). *At the theoretical limit of sustainability [1, 26], the maximum neural drive and duty cycle are:*

$$C_{\max} = \frac{FR}{F+R}\ [\text{min}^{-1}], \qquad \delta_{\max} = \frac{R}{F+R}\ [\textit{dimensionless}]. \tag{6}$$

Proof. At the sustainability limit [1], the resting pool is fully depleted: $M_R = 0$, hence $M_A + M_F = 1$. Setting all derivatives to zero in the work phase ($R_{\text{eff}} = R$):
Step 1. From (2) at steady state:

$$C - F \cdot M_A = 0 \implies M_A = \frac{C}{F}. \tag{7}$$

Step 2. From (3) at steady state:

$$F \cdot M_A - R \cdot M_F = 0 \implies M_F = \frac{F \cdot M_A}{R} = \frac{C}{R}. \tag{8}$$

Step 3. Substituting into $M_A + M_F = 1$:

$$\frac{C}{F} + \frac{C}{R} = 1 \implies C\left(\frac{R+F}{FR}\right) = 1 \implies C = \frac{FR}{F+R} = C_{\max}.$$

Step 4. The duty cycle $\delta = M_A$ at steady state. From (7):

$$\delta_{\max} = \frac{C_{\max}}{F} = \frac{1}{F} \cdot \frac{FR}{F+R} = \frac{R}{F+R}.$$

This matches Liu et al. [1], Eq. 12, and Xia & Frey-Law [2], Eq. 7. □

Remark 3.5 (Dimensional Check). $C_{\max} = FR/(F+R)$ *has units* $[\text{min}^{-1}][\text{min}^{-1}]/[\text{min}^{-1}] = \text{min}^{-1}$, *consistent with a rate.* $\delta_{\max} = R/(F+R)$ *is* $[\text{min}^{-1}]/[\text{min}^{-1}]$, *correctly dimensionless.*

3.5 Calibrated Parameters

Parameters F and R are from Frey-Law, Looft & Heitsman [3] (Monte Carlo calibration against the 194-publication meta-analysis of Frey-Law & Avin [6], itself building on the foundational endurance curves of Rohmert [26]). Reperfusion multiplier r values from Looft et al. [4], with shoulder validated by [5].

Table 1: Calibrated 3CC-r parameters. $\delta_{\max} = R/(F+R)$; rest-phase threshold $\Theta_{\max}^{\min} = F/(F+Rr)$.

Muscle Group	F [3]	R [3]	r [4, 5]	$\delta_{\max}$	$\Theta_{\max}^{\min}$	Note [6]
Shoulder (overhead)	0.0146	0.00058	15	3.8%	62.7%	Highest threshold
Ankle (walking)	0.00589	0.0182	15	75.5%	2.1%	$Rr \gg F$
Knee (lifting)	0.0150	0.00175	15	10.4%	36.4%	Heavy load
Elbow (flexion)	0.00912	0.00094	15	9.3%	39.3%	Moderate
Trunk (extension)	0.00657	0.00354	15	35.0%	11.0%	Postural
Hand/Grip (squeeze)	0.00794	0.00109	30	12.1%	19.5%	Fine motor

Verification. Shoulder: $0.00058/(0.0146 + 0.00058) = 0.00058/0.01518 = 0.0382 = 3.8\%$. ✓
Ankle: $0.0182/(0.00589 + 0.0182) = 0.0182/0.02409 = 0.7554 = 75.5\%$. ✓

The fatigue-resistance ranking (ankle > trunk > grip > knee > elbow > shoulder) matches [6, 26].

4 Constraint Inference: MMICRL

To learn individual worker safety constraints from heterogeneous demonstrations, we use Multi-Modal Inverse Constrained Reinforcement Learning (MMICRL) [8]. MMICRL builds on Maximum Entropy IRL [7], Inverse Constrained RL [9], and multi-modal latent variable imitation [10], operating within the Constrained Markov Decision Process (CMDP) framework [24] as applied to safe policy optimisation by Achiam et al. [11].

4.1 Objective Function (Corrected)

Definition 4.1 (MMICRL Weighted Objective)**.** *Let z denote a latent variable representing agent type, and τ a trajectory. The MMICRL objective is:*

$$\max_{\pi} \left\{\lambda_1 \, H[\pi(\tau)] - \lambda_2 \, H[\pi(\tau \mid z)]\right\}, \tag{9}$$

where:

- $H[\pi(\tau)]$*: marginal entropy of the policy over trajectories [7]. Maximised to ensure behavioural diversity across all agent types.*
- $H[\pi(\tau \mid z)]$*: conditional entropy of trajectories given agent type [8, 10]. Minimised (via the negative sign) to ensure behavioural consistency within each type.*
- $\lambda_1, \lambda_2 > 0$*: weighting coefficients controlling the exploration–specialisation trade-off [8].*

Theorem 4.2 (Decomposition of the Weighted Objective)**.** *The objective* (9) *decomposes as:*

$$\lambda_1 \, H[\pi(\tau)] - \lambda_2 \, H[\pi(\tau \mid z)] = \lambda_2 \, I(\tau; z) + (\lambda_1 - \lambda_2) \, H[\pi(\tau)], \tag{10}$$

where $I(\tau; z) = H[\pi(\tau)] - H[\pi(\tau \mid z)]$ is the mutual information between trajectories and agent types.

Proof. Starting from the definition of mutual information:

$$H[\pi(\tau)] = I(\tau; z) + H[\pi(\tau \mid z)].$$

Substituting $H[\pi(\tau \mid z)] = H[\pi(\tau)] - I(\tau; z)$ into (9):

$$\begin{aligned} \lambda_1 H[\pi(\tau)] - \lambda_2 \big(H[\pi(\tau)] - I(\tau; z)\big) &= \lambda_1 H[\pi(\tau)] - \lambda_2 H[\pi(\tau)] + \lambda_2 I(\tau; z) \\ &= \lambda_2 \, I(\tau; z) + (\lambda_1 - \lambda_2) \, H[\pi(\tau)]. \end{aligned}$$

□

Remark 4.3 (Interpretation and the Mutual Information Claim)**.** *Previous versions claimed* (9) *was "equivalent to maximising mutual information $I(\tau; z)$." This is correct* ***only when*** *$\lambda_1 = \lambda_2$, in which case the residual $(\lambda_1 - \lambda_2) H[\pi(\tau)]$ vanishes.*

When $\lambda_1 \neq \lambda_2$, the objective jointly optimises two terms:

1. $\lambda_2 \, I(\tau; z)$*: mutual information, promoting between-type separability and within-type coherence.*
2. $(\lambda_1 - \lambda_2) \, H[\pi(\tau)]$*: a marginal entropy bonus (if $\lambda_1 > \lambda_2$) or penalty (if $\lambda_1 < \lambda_2$).*

Case analysis:

- $\lambda_1 > \lambda_2$*: The objective rewards overall policy stochasticity beyond what mutual information alone requires. This encourages broader state-space coverage but may introduce within-type variability.*

- $\lambda_1 < \lambda_2$: *The objective penalises marginal entropy, preferring compact trajectory distributions. Risk of mode collapse if the penalty is too strong.*
- $\lambda_1 = \lambda_2$: *Pure mutual information maximisation. This is the recommended default for HC-MARL unless domain-specific reasons dictate otherwise.*

Remark 4.4 (Operational Recommendation). *For HC-MARL, we recommend setting $\lambda_1 = \lambda_2 = \lambda$, yielding:*

$$\max_{\pi} \; \lambda \, I(\tau; z), \tag{11}$$

under which the mutual information equivalence holds exactly. This maximises between-type diversity and within-type consistency without introducing spurious entropy biases. If $\lambda_1 \neq \lambda_2$ is used, the residual term must be reported and its effect on learned constraints analysed.

5 Safety Filters: Ergonomic Control Barrier Functions

Safety verification via barrier certificates originates with Prajna & Jadbabaie [16]. Ames et al. [14, 15] formalised Control Barrier Functions (CBFs) for real-time safety-critical control. Standard CBFs are insufficient here because the control input $C(t)$ does not appear in the first derivative of the fatigue constraint (relative degree > 1 [25]), requiring Exponential CBFs [12] generalised by Xiao & Belta [13].

5.1 Fatigue Ceiling Barrier (Primary)

Definition 5.1 (Fatigue Barrier). *Define the primary barrier function:*

$$h(x) = \Theta_{\max} - M_F \geq 0, \tag{12}$$

where $\Theta_{\max} \in (0, 1)$ is the maximum allowable fatigue fraction.

First derivative. Using (3):

$$\dot{h} = -\frac{dM_F}{dt} = -F \cdot M_A + R_{\text{eff}} \cdot M_F. \tag{13}$$

The control input $C(t)$ does not appear. Hence relative degree ≥ 2 [25, 12].
Second derivative. Differentiating (13) and substituting the ODEs (2)–(3):

$$\ddot{h} = -F\frac{dM_A}{dt} + R_{\text{eff}}\frac{dM_F}{dt} = -F\big(C - FM_A\big) + R_{\text{eff}}\big(FM_A - R_{\text{eff}}M_F\big) = -FC + F^2 M_A + R_{\text{eff}} F M_A - R_{\text{eff}}^2 M_F. \tag{14}$$

The input $C(t)$ appears with coefficient $-F$. Since $F > 0$, the system has relative degree 2 with respect to h [12, 13, 25].

5.2 ECBF Construction

Following Nguyen & Sreenath [12], define the composite barriers:

$$\psi_0(x) = h(x), \tag{15}$$

$$\psi_1(x) = \dot{h} + \alpha_1 \, h(x), \tag{16}$$

where $\alpha_1 > 0$ is a class-$\mathcal{K}$ design parameter [25, 12]. The ECBF condition enforces [12, 13]:

$$\dot{\psi}_1(x, C) \geq -\alpha_2 \, \psi_1(x), \tag{17}$$

where $\alpha_2 > 0$. Expanding $\dot{\psi}_1 = \ddot{h} + \alpha_1 \dot{h}$ and substituting (13)–(14):

$$\underbrace{-FC}_{\text{control}} + \underbrace{F^2 M_A + R_{\text{eff}} F M_A - R_{\text{eff}}^2 M_F + \alpha_1(-F M_A + R_{\text{eff}} M_F)}_{\text{state-dependent}} \geq -\alpha_2 \underbrace{\left(-F M_A + R_{\text{eff}} M_F + \alpha_1(\Theta_{\max} - M_F)\right)}_{\psi_1}. \tag{18}$$

Solving for $C(t)$ (since the coefficient of C is $-F < 0$, the inequality flips):

$$C(t) \leq \frac{1}{F}\Big[F^2 M_A + R_{\text{eff}} F M_A - R_{\text{eff}}^2 M_F + \alpha_1(-F M_A + R_{\text{eff}} M_F) + \alpha_2\big(-F M_A + R_{\text{eff}} M_F + \alpha_1(\Theta_{\max} - M_F)\big)\Big]. \tag{19}$$

This is implemented as a CBF-QP [14]: given the nominal RL action C_{nom}, solve:

$$C^*(t) = \arg\min_C \ \|C - C_{\text{nom}}\|^2 \quad \text{s.t. } C \text{ satisfies (18) and (23) and } C \geq 0. \tag{20}$$

5.3 Resting Floor Barrier (New)

Definition 5.2 (Resting Pool Barrier). *To prevent the physiologically meaningless state $M_R < 0$, define:*

$$h_2(x) = M_R = 1 - M_A - M_F \geq 0. \tag{21}$$

First derivative. Using (4):

$$\dot{h}_2 = \frac{dM_R}{dt} = R_{\text{eff}} \cdot M_F - C(t). \tag{22}$$

The input $C(t)$ appears directly with coefficient -1. Hence h_2 has relative degree 1 [25, 14].

Since h_2 has relative degree 1, a standard CBF suffices [14, 15]:

$$\dot{h}_2 \geq -\alpha_3 h_2 \implies C(t) \leq R_{\text{eff}} \cdot M_F + \alpha_3(1 - M_A - M_F), \tag{23}$$

where $\alpha_3 > 0$ is a class-$\mathcal{K}$ design parameter [25]. This ensures that as $M_R \to 0$, the maximum allowable neural drive is progressively restricted, preventing over-recruitment from an empty resting pool.

Remark 5.3 (Dual Barrier QP). *The CBF-QP* (20) *now includes two constraints:* (18) *(fatigue ceiling, relative degree 2) and* (23) *(resting floor, relative degree 1). Both are linear in C, so the QP remains convex [27] and efficiently solvable via CVXPY/OSQP [28].*

5.4 Safety Across Mode Transitions

The 3CC-r system [4] switches between $R_{\text{eff}} = R$ (work) and $R_{\text{eff}} = Rr$ (rest). The ECBF (Sections 5.1–5.3) guarantees $h \geq 0$ and $h_2 \geq 0$ during work by filtering $C(t)$. During rest, $C = 0$ and no filter is active. We must prove that the autonomous rest-phase dynamics preserve $h(x) = \Theta_{\max} - M_F \geq 0$.

5.4.1 Rest-Phase Fatigue Overshoot

When a worker stops exerting ($C \to 0$), the currently active motor units $M_A > 0$ continue to transition into the fatigued pool at rate F (metabolic fatigue does not cease instantaneously [1, 4]). If the reperfusion-enhanced recovery rate Rr is small relative to F, the fatigued fraction M_F can increase transiently before it decreases. Quantitatively, from the rest-phase ODE:

$$\left.\frac{dM_F}{dt}\right|_{C=0} = F \cdot M_A - Rr \cdot M_F. \tag{24}$$

This is positive whenever $M_A/M_F > Rr/F$. For the shoulder ($Rr/F = 0.596$), this condition is easily met during normal work, confirming that fatigue can rise during early rest.

Therefore, a proof of rest-phase safety must account for this overshoot explicitly.

5.4.2 Design Requirement and Rest-Phase Safety Proof

Definition 5.4 (Rest-Phase Safety Threshold)**.** *For a muscle group with parameters (F, R, r), define the rest-phase safety threshold:*

$$\Theta_{\max}^{\min} = \frac{F}{F + Rr}. \tag{25}$$

Assumption 5.5 (Design Requirement)**.** *For each muscle group, the fatigue ceiling satisfies:*

$$\Theta_{\max} \geq \Theta_{\max}^{\min} = \frac{F}{F + Rr}. \tag{26}$$

Table 2: Rest-phase parameters. Rr/F determines whether fatigue can transiently increase during rest. $\Theta_{\max}^{\min} = F/(F + Rr)$ is the minimum safety threshold that guarantees rest-phase invariance (Theorem 5.7).

Muscle Group	Rr [min^{-1}]	Rr/F	Overshoot?	$\Theta_{\max}^{\min}$
Shoulder	0.00870	0.596	Yes (< 1)	62.7%
Ankle	0.27300	46.35	No	2.1%
Knee	0.02625	1.750	Marginal	36.4%
Elbow	0.01410	1.546	Marginal	39.3%
Trunk	0.05310	8.082	No	11.0%
Grip	0.03270	4.118	Unlikely	19.5%

Remark 5.6. *Assumption 5.5 states that the safety threshold must not be set below the maximum fatigue that could be reached from a fully depleted resting pool during autonomous rest. For the most restrictive muscle group (shoulder), this requires $\Theta_{\max} \geq 62.7\%$. This is a reasonable operational constraint: allowing a shoulder to fatigue up to 63% MVC is consistent with ergonomic guidelines for heavy intermittent work [26, 6]. For all other muscle groups, the requirement is milder (Table 2).*

Theorem 5.7 (Rest-Phase Safety)**.** *Under Assumption 5.5, if $h(x(t_s)) = \Theta_{\max} - M_F(t_s) \geq 0$ and $h_2(x(t_s)) = M_R(t_s) \geq 0$ at any work-to-rest transition time t_s, then $h(x(t)) \geq 0$ for all $t \geq t_s$ during the rest phase.*

Proof. During rest, $C(t) = 0$ and $R_{\text{eff}} = Rr$. The dynamics are autonomous.
Step 1: M_R is non-decreasing during rest. From (4) with $C = 0$:

$$\frac{dM_R}{dt} = Rr \cdot M_F \geq 0.$$

Hence $M_R(t) \geq M_R(t_s) \geq 0$ for all $t \geq t_s$.
Step 2: Nagumo invariance at the boundary $M_F = \Theta_{\max}$. Suppose the trajectory reaches the boundary $M_F(t_1) = \Theta_{\max}$ at some time $t_1 \geq t_s$. By conservation (1):

$$M_A(t_1) = 1 - M_R(t_1) - \Theta_{\max}.$$

Substituting into (24):

$$\left.\frac{dM_F}{dt}\right|_{t_1} = F\big(1 - M_R(t_1) - \Theta_{\max}\big) - Rr \cdot \Theta_{\max} = F\big(1 - M_R(t_1)\big) - (F + Rr)\,\Theta_{\max}. \tag{27}$$

By Assumption 5.5, $\Theta_{\max} \geq F/(F+Rr)$, so $(F+Rr)\Theta_{\max} \geq F$. Since $M_R(t_1) \geq 0$ (from Step 1):

$$\left.\frac{dM_F}{dt}\right|_{t_1} \leq F - (F+Rr)\Theta_{\max} \leq 0.$$

Equivalently, $\dot{h}(t_1) = -dM_F/dt|_{t_1} \geq 0$: the fatigue barrier is non-decreasing at the boundary.

Step 3: Conclusion by Nagumo's theorem. The set $\mathcal{S} = \{x : M_F \leq \Theta_{\max}\}$ satisfies the subtangentiality condition: at every boundary point $M_F = \Theta_{\max}$, the velocity field points inward ($dM_F/dt \leq 0$). By Nagumo's invariance theorem [22] (see also Blanchini [23] for a modern treatment in control), $\mathcal{S}$ is forward-invariant under the rest-phase dynamics. Since $M_F(t_s) \leq \Theta_{\max}$, we have $M_F(t) \leq \Theta_{\max}$ for all $t \geq t_s$. □

Remark 5.8 (Tightness of the Threshold)**.** *The bound $\Theta_{\max}^{\min} = F/(F+Rr)$ is tight. At $M_R = 0$, $M_A = 1 - \Theta_{\max}$, and $M_F = \Theta_{\max} = F/(F+Rr)$:*

$$\frac{dM_F}{dt} = F(1-\Theta_{\max}) - Rr\,\Theta_{\max} = \frac{F \cdot Rr}{F+Rr} - \frac{Rr \cdot F}{F+Rr} = 0.$$

So M_F is exactly at an equilibrium. If $\Theta_{\max}$ were set below $F/(F+Rr)$, then $dM_F/dt > 0$ at the boundary, and the safe set would be violated.

5.4.3 Work-to-Rest Transition

Proposition 5.9 (Positive ψ_1 Jump)**.** *At a work-to-rest switching instant t_s, the composite barrier ψ_1 experiences a positive jump:*

$$\psi_1^{\text{rest}}(t_s) - \psi_1^{\text{work}}(t_s) = R(r-1)M_F(t_s) > 0. \tag{28}$$

Proof. $\psi_1 = \dot{h} + \alpha_1 h = (-FM_A + R_{\text{eff}}M_F) + \alpha_1(\Theta_{\max} - M_F)$. Since h and M_A are continuous but R_{eff} jumps from R to Rr:

$$\psi_1^{\text{rest}} - \psi_1^{\text{work}} = (Rr - R)M_F(t_s) = R(r-1)M_F(t_s) > 0,$$

because $r > 1$ and $M_F(t_s) > 0$ (the worker was active). Thus the ECBF feasibility condition $\psi_1 \geq 0$ is strengthened at the transition to rest. □

5.4.4 Rest-to-Work Transition

Proposition 5.10 (ECBF Feasibility at Resumption)**.** *After a rest period of sufficient duration, $\psi_1^{\text{work}}(t_r) > 0$ at the rest-to-work transition, and the ECBF is feasible when work resumes. The minimum rest duration required satisfies:*

$$\Delta t^* = \left(\frac{1}{F}\ln\frac{FM_A(t_s)}{\beta(t_r)}\right)^+, \tag{29}$$

where $(\cdot)^+ = \max(\cdot, 0)$, $\beta(t_r) = R\,M_F(t_r) + \alpha_1(\Theta_{\max} - M_F(t_r))$, and $t_r = t_s + \Delta t$ is the resumption time. If $\Delta t < \Delta t^*$, then $\psi_1^{\text{work}}(t_r) < 0$; the CBF-QP restricts $C^*(t_r)$ to a small value (or zero), and the ECBF condition drives ψ_1 back toward non-negative values. Safety ($M_F \leq \Theta_{\max}$) is maintained unconditionally by Theorem 5.7.

Proof. At the rest-to-work transition at time $t_r > t_s$, R_{eff} drops from Rr to R, causing ψ_1 to decrease by $R(r-1)M_F(t_r)$. The post-transition value is:

$$\psi_1^{\text{work}}(t_r) = -FM_A(t_r) + RM_F(t_r) + \alpha_1(\Theta_{\max} - M_F(t_r)).$$

During rest, M_A decays exponentially [1]: $M_A(t) = M_A(t_s)e^{-F(t-t_s)}$. Define $\beta(t_r) = RM_F(t_r) + \alpha_1(\Theta_{\max} - M_F(t_r)) > 0$ (positive since $M_F(t_r) \leq \Theta_{\max}$ by Theorem 5.7 and $R, \alpha_1 > 0$). Then:

$$\psi_1^{\text{work}}(t_r) = \beta(t_r) - FM_A(t_s)e^{-F\Delta t}.$$

Setting $\psi_1^{\text{work}}(t_r) = 0$ and solving:

$$\Delta t^* = \frac{1}{F} \ln \frac{FM_A(t_s)}{\beta(t_r)},$$

which is positive when $FM_A(t_s) > \beta(t_r)$ (true for muscle groups with $Rr/F < 1$, e.g., shoulder). For $\Delta t > \Delta t^*$, $\psi_1^{\text{work}}(t_r) > 0$ and the ECBF is feasible.

Behaviour when $\Delta t < \Delta t^*$ (early resumption). If the scheduler attempts to resume work before Δt^*, we have $\psi_1 < 0$. The ECBF condition (17) requires $\dot{\psi}_1 \geq -\alpha_2\psi_1 > 0$, which imposes a strict upper bound on C via (18). Whether this bound forces $C^* = 0$ or permits a small positive C^* depends on the state and on the gains α_1, α_2 [12, 13]:

- **Sufficiently large α_2:** The ECBF upper bound on C is non-positive whenever $\psi_1 < 0$, forcing $C^* = 0$ (mandatory rest). The QP automatically extends the rest period.
- **Small α_1, α_2:** The ECBF upper bound may permit a small positive C^*. In this regime, ψ_1 recovers toward non-negative values through a combination of restricted (low-intensity) work and continued decay of M_A.

In both cases, the ECBF condition $\dot{\psi}_1 \geq -\alpha_2\psi_1$ enforces monotonic recovery of ψ_1 toward the feasible region [12]. Crucially, the primary safety guarantee $M_F \leq \Theta_{\max}$ holds *unconditionally* by Theorem 5.7 (Nagumo invariance [22, 23]), independently of the ECBF composite barrier state.

Asymptotic guarantee. As $\Delta t \to \infty$, $M_A(t_r) \to 0$ and $M_F(t_r) \to 0$, so $\psi_1^{\text{work}} \to \alpha_1\Theta_{\max} > 0$. ECBF feasibility is always eventually restored. □

Remark 5.11 (Implicit Nature of Δt^*)**.** *Equation* (29) *is implicit: $\beta(t_r)$ depends on $M_F(t_r)$, which evolves during rest and thus depends on Δt itself. A conservative (worst-case) upper bound on Δt^* is obtained by evaluating β at its minimum over $M_F \in [0, \Theta_{\max}]$. Since $\beta = (R - \alpha_1)M_F + \alpha_1\Theta_{\max}$, the minimum occurs at $M_F = \Theta_{\max}$ when $\alpha_1 \geq R$ and at $M_F = 0$ when $\alpha_1 < R$, giving $\beta_{\min} = \min(R, \alpha_1)\,\Theta_{\max}$. Hence:*

$$\Delta t^* \leq \overline{\Delta t} = \left(\frac{1}{F} \ln \frac{FM_A(t_s)}{\min(R, \alpha_1)\,\Theta_{\max}} \right)^+. \tag{30}$$

This closed-form bound is sufficient for implementation: resting for $\overline{\Delta t}$ guarantees $\psi_1 \geq 0$ at resumption.

Remark 5.12 (Numerical Example)**.** *For the shoulder ($F = 0.0146$, $R = 0.00058$, $Rr/F = 0.596$) with the recommended $\alpha_1 = 0.05 > R$, so $\min(R, \alpha_1) = R$ and the bound reduces to $\overline{\Delta t} = (F^{-1}\ln(FM_A(t_s)/(R\Theta_{\max})))^+$. With $M_A(t_s) = \delta_{\max} = 3.8\%$ and M_F near $\Theta_{\max}$: $\overline{\Delta t} \approx$ 29 minutes. For larger initial $M_A = 0.35$ (extreme case): $\overline{\Delta t} \approx 3$ hours. For muscle groups with $Rr/F \gg 1$ (e.g., ankle, trunk), $\Delta t^* = 0$ and $\psi_1^{\text{work}} > 0$ immediately.*

5.4.5 QP Feasibility and Automatic Rest

Remark 5.13 (Mandatory Rest from QP Infeasibility)**.** *The CBF-QP* (20) *imposes two upper bounds on $C(t)$—from* (18) *and* (23)*—plus the constraint $C \geq 0$. When the physiological state is severely degraded (high M_F, low M_R), both upper bounds may become non-positive, forcing $C^*(t) = 0$ (mandatory rest). This is a desirable feature: the QP [27, 28] automatically mandates rest when no positive neural drive is safe, providing a third independent safety mechanism alongside the ECBF barrier and the NSWF disagreement utility (Section 6).*

6 Cooperative Task Allocation: Nash Social Welfare

6.1 Terminological Correction

The optimisation in this section is solved by a central planner, not by agents bargaining with each other. The correct framing is the **Nash Social Welfare Function (NSWF)**, which maximises the product of agent surpluses. The NSWF is axiomatically connected to Nash's bargaining theory [17]: Kaneko & Nakamura [21] showed that the NSWF is the unique allocation rule satisfying Pareto optimality, symmetry, and independence of irrelevant alternatives for N-player settings, exactly extending Nash's 2-player axioms [17]. The disagreement point formulation draws on Nash's endogenous threat model [18] and the outside option principle of Binmore, Shaked & Sutton [20]. The log-transform for gradient-based computation follows Navon et al. [19].

6.2 Definitions

Definition 6.1 (Task Assignment). *Let $\mathcal{I} = \{1, \ldots, N\}$ be the set of workers and $\mathcal{J} = \{1, \ldots, M\}$ the set of productive tasks available in the current allocation round. The augmented task set includes a rest option:*

$$\mathcal{J}_0 = \mathcal{J} \cup \{0\}, \quad \textit{where task 0 denotes "rest (no work)."}$$

A binary assignment $a_{ij} = 1$ if task $j \in \mathcal{J}_0$ is assigned to worker i, and $a_{ij} = 0$ otherwise, subject to:

(i) Each worker receives exactly one assignment: $\sum_{j \in \mathcal{J}_0} a_{ij} = 1$ for all i.

(ii) Each productive task is assigned to at most one worker: $\sum_{i \in \mathcal{I}} a_{ij} \leq 1$ for all $j \in \mathcal{J}$.

(iii) Multiple workers may rest simultaneously: no capacity constraint on task 0.

Let $U(i, j)$ denote the productivity utility of worker i performing task $j \in \mathcal{J}$. The rest option yields:

$$U(i, 0) = D_i(M_F^i) + \epsilon, \quad \epsilon > 0 \textit{ (small)}, \tag{31}$$

so that resting produces a surplus of exactly ϵ, ensuring the log argument remains well-defined.

Definition 6.2 (Divergent Disagreement Utility). *The disagreement utility D_i represents the value of the outside option (resting) for worker i [17, 20]:*

$$D_i(M_F^i) = \kappa \cdot \frac{(M_F^i)^2}{1 - M_F^i}, \quad \kappa > 0. \tag{32}$$

Proposition 6.3 (Properties of D_i). *The disagreement utility* (32) *satisfies:*

(P1) $D_i(0) = 0$ *(no premium when fresh).*

(P2) $D_i'(M_F) > 0$ *for* $M_F \in (0, 1)$ *(monotonically increasing).*

(P3) $D_i(M_F) \to +\infty$ *as* $M_F \to 1^-$ *(infinite cost at burnout).*

Proof. **(P1):** $D_i(0) = \kappa \cdot 0/1 = 0$. ✓
(P2): Differentiating (32):

$$D_i'(M_F) = \kappa \cdot \frac{d}{dM_F}\left(\frac{(M_F)^2}{1 - M_F}\right) = \kappa \cdot \frac{2M_F(1 - M_F) + (M_F)^2}{(1 - M_F)^2} = \kappa \cdot \frac{M_F(2 - M_F)}{(1 - M_F)^2}.$$

For $M_F \in (0, 1)$: $M_F > 0$, $(2 - M_F) > 0$, $(1 - M_F)^2 > 0$, so $D_i' > 0$. ✓
(P3): As $M_F \to 1^-$, the numerator $(M_F)^2 \to 1 > 0$ and the denominator $(1 - M_F) \to 0^+$, so $D_i \to +\infty$. ✓ □

Remark 6.4 (Low-Fatigue Regime). *For $M_F \ll 1$: $(1 - M_F)^{-1} \approx 1$, so $D_i \approx \kappa\,(M_F)^2$. The disagreement utility recovers smooth quadratic sensitivity during routine operations, with the divergence activating only near the physiological limit.*

6.3 Allocation Objective

Definition 6.5 (Nash Social Welfare Objective). *The central planner solves:*

$$\max_{\{a_{ij}\}} \sum_{i=1}^{N} \ln\big(U(i, j^*(i)) - D_i(M_F^i)\big), \tag{33}$$

where $j^*(i) \in \mathcal{J}_0$ denotes the task assigned to worker i, subject to constraints (i)–(iii) in Definition 6.1 and the surplus $U(i, j) - D_i(M_F^i) > 0$ for every assignment (guaranteed for rest by (31)*).*

Dynamics. If worker i is fatigued (M_F^i high), then D_i is large [20]. For the surplus $U(i, j) - D_i$ of a productive task to remain positive, the utility must exceed the fatigue premium $\kappa(M_F^i)^2/(1-M_F^i)$ [17, 18]. When no productive task can pay this premium, the planner assigns worker i to rest (task 0), contributing $\ln(\epsilon) \ll 0$ to the objective—strongly disfavoured but feasible. This handles all partial assignment scenarios: more workers than tasks ($N > M$), more tasks than workers ($N < M$), or individual workers too fatigued for any productive assignment.

As $M_F^i \to 1$, no finite $U(i, j)$ can overcome $D_i \to \infty$. This provides a mathematical guarantee against burnout, independent of the safety filter.

Remark 6.6 (Relationship to Nash Bargaining). *For $N = 2$ with symmetric information, the NSWF solution coincides with the Nash Bargaining Solution [17]. For $N > 2$, the NSWF remains the unique rule satisfying the multi-player generalisation of Nash's axioms [21]. The use of "Nash Social Welfare" rather than "Nash Bargaining" is preferred for $N > 2$ centralised settings to avoid conflating the welfare function with the 2-player cooperative game concept.*

7 Action-to-Neural-Drive Interface

The four preceding modules operate at two levels of abstraction:

- **Allocation layer** (NSWF, Section 6): Assigns discrete task j to worker i.
- **Physiological layer** (3CC-r, ECBF, Sections 3–5): Operates on continuous neural drive $C(t)$.

The interface mapping connects these layers.

7.1 Task-to-Load Mapping

Each task $j \in \mathcal{J}$ is characterised by a demand profile: a set of target loads across relevant muscle groups.

Definition 7.1 (Task Demand Profile). *For task j involving muscle group g, define:*

$$T_L^{(j)}{}_g \in [0, 1], \tag{34}$$

the fraction of Maximum Voluntary Contraction (MVC) required from muscle group g during task j. Tasks requiring multiple muscle groups have a vector of demands $T_L^{(j)} = (T_{L,g_1}^{(j)}, \ldots, T_{L,g_K}^{(j)})$.

7.2 Neural Drive Controller

Following the feedback controller structure of Xia & Frey-Law [2], the neural drive $C(t)$ is determined by a feedback law that recruits motor units to meet the target load:

$$C(t) = \begin{cases} k_p\big(T_L(t) - M_A(t)\big)^+ & \text{if assigned (work phase)}, \\ 0 & \text{if resting}, \end{cases} \tag{35}$$

where $k_p > 0$ is a proportional gain, and $(\cdot)^+ = \max(\cdot, 0)$ ensures non-negative recruitment. The controller attempts to maintain $M_A(t) \approx T_L(t)$ by recruiting from the resting pool when the active fraction falls below the target.

Remark 7.2. *In HC-MARL, the RL agent replaces the proportional controller with a learned policy $\pi_\theta(C \mid x, \text{task})$ that outputs a continuous neural drive. The feedback law* (35) *serves as both the baseline controller and the behavioural prior from which MMICRL demonstrations are generated. The ECBF safety filter* (20) *then clips any proposed $C(t)$ to the safe set, regardless of whether it comes from* (35) *or the RL policy.*

7.3 End-to-End Pipeline

The complete pipeline for each allocation round proceeds as:

1. **State observation.** Read $x_i(t) = [M_R^i, M_A^i, M_F^i]$ for each worker i.
2. **Task allocation.** Central planner solves (33) to assign tasks $\{j^*(i)\}$.
3. **Load translation.** For each assigned worker–task pair $(i, j^*(i))$, look up ${T_L^{(j^*(i))}}_g$.
4. **Neural drive.** RL policy (or baseline (35)) proposes $C_{\text{nom}}(t)$.
5. **Safety filtering.** CBF-QP (20) clips $C_{\text{nom}}(t) \to C^*(t)$.
6. **State update.** Integrate ODEs (2)–(4) with $C^*(t)$.
7. **Repeat.** Return to step 1 at the next allocation interval.

8 Summary of All Corrections

#	Issue	Previous Version	This Version
1	MMICRL–MI equivalence	Claimed $\lambda_1 H - \lambda_2 H(\cdot\|z) = I(\tau; z)$ for all λ_1, λ_2	Proved decomposition; equivalence holds iff $\lambda_1 = \lambda_2$; recommended $\lambda_1 = \lambda_2$
2	Switched-mode safety	Used relative-degree-1 argument ($\dot{h}$ comparison); claimed $Rr/F \gg 1$ for all muscles (false for shoulder: $Rr/F = 0.60$)	Nagumo invariance proof [22, 23] with explicit design requirement $\Theta_{\max} \geq F/(F + Rr)$; numerical thresholds for all 6 muscle groups
3	Resting pool constraint	Not addressed; only $M_F \leq \Theta_{\max}$ guarded	Added $h_2 = M_R \geq 0$ as relative-degree-1 CBF [14] in dual-barrier QP [27]
4	"Nash Bargaining"	Called centralised N-player allocation "Nash Bargaining"	Correctly identified as Nash Social Welfare Function [21]; axiomatic connection to Nash [17] preserved
5	Action-to-$C(t)$ mapping	Absent	New Section 7: task demand profiles, neural drive controller [2], end-to-end pipeline
6	Assignment notation	τ_{ij}: undefined subscript; symbol overloaded with trajectory τ (Section 4)	$a_{ij} \in \{0, 1\}$: standard assignment variable; τ reserved exclusively for trajectories
7	Utility notation	$U_{\text{work}}(\tau_{ij})$: utility of a binary indicator (semantically ill-typed)	$U(i, j)$: utility depends on the worker–task pair

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